

Backend Service Documentation Template

1. Introduction

1.1. Purpose

[Describe the purpose of this backend service documentation.]

1.2. Scope

[Define the scope of the backend service.]

1.3. Service Overview

[Provide a high-level overview of the backend service.]

2. System Architecture

2.1. Architecture Summary

[Describe backend architecture and major components.]

2.2. Technology Stack

Component	Technology
Backend Framework	[e.g., FastAPI / Express.js / Spring Boot]
Database	[e.g., PostgreSQL / MongoDB]
Authentication	[e.g., JWT / OAuth2]
Cloud Provider	[e.g., AWS / Azure / GCP]

3. Service Configuration

Configuration Item	Description	Default Value
PORT	[Application Port]	[8000]
DATABASE_URL	[Database Connection String]	[Value]
JWT_SECRET	[Authentication Secret]	[Value]

4. API Endpoints

Endpoint	Method	Description	Authentication Required
/api/example	GET	[Describe endpoint purpose]	Yes/No

5. Request and Response Structure

Field	Type	Description	Required
id	String	[Field Description]	Yes
name	String	[Field Description]	Yes

5.1. Sample Request

```
{
  "field": "value"
}
```

5.2. Sample Response

```
{
  "success": true,
  "data": {}
}
```

6. Authentication and Authorization

6.1. Authentication Method

[Describe authentication mechanism.]

6.2. Roles and Permissions

Role	Permissions
Admin	[Full Access]
User	[Limited Access]

7. Database Design

Table / Collection	Purpose
users	[Stores user information]
logs	[Stores application logs]

8. Error Handling

Error Code	Description	HTTP Status
AUTH_001	Unauthorized Access	401
SERVER_001	Internal Server Error	500

9. Logging and Monitoring

9.1. Logging Strategy

[Describe application logging strategy.]

9.2. Monitoring Tools

Tool	Purpose
Prometheus	[Metrics Monitoring]
Grafana	[Visualization Dashboard]

10. Security Considerations

Security Area	Implementation
Authentication	[JWT / OAuth2]
Encryption	[TLS / AES-256]
Rate Limiting	[Enabled / Disabled]

11. Deployment Details

Environment	URL / Details
Development	[Dev Environment URL]
Production	[Production URL]

11.1. Deployment Workflow

[Describe CI/CD and deployment process.]

12. Dependencies

Dependency	Purpose	Version
FastAPI	[Backend Framework]	[Version]
SQLAlchemy	[ORM]	[Version]

13. Conclusion

[Provide final notes and recommendations.]